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REG NO: I231/G/4468/22

TAKE AWAY CAT TWO

COURSE CODE: ITB 14131

COURSE TITLE: GROUP DESIGN WORKSHOP

1. **You have been appointed as a project manager to lead the development of a new software application for a client operating in the e-commerce sector. The client has defined clear requirements for the project, including strict time and budget constraints. The primary purpose of the software application is to improve the overall user experience of the client's ecommerce platform. The project team is made up of software developers, quality assurance specialists, and a user interface designer.**

   I) Identify and prioritize three major project objectives that are consistent with the client's expectations and the goal of improving user experience on the e-commerce platform.

E-commerce Platform Enhancement Project

I Prioritized Project Objectives

1. **Optimize Core User Journeys:** Streamline product discovery, cart management, and checkout processes to reduce steps by 40% and decrease cart abandonment rate by 25% within 6 months.
2. **Enhance Mobile Experience:** Achieve 100% mobile responsiveness with page load times under 2 seconds on 3G networks, targeting a 30% increase in mobile conversion rates within 4 months.
3. **Personalize User Engagement:** Implement AI-driven product recommendations and dynamic content customization to increase average order value by 15% and customer retention by 20% within 8 months.

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Ii SMART Framework Implementation

Specific: Each objective defines exact processes to improve (e.g.checkout flow not "user experience"). Importance: Prevents scope ambiguity in cross-functional teams.

Measurable: Includes quantifiable metrics (e.g.25% reduction in cart abandonment). Importance: Enables objective progress tracking against budget/time constraints.

Achievable: Based on team capacity analysis and phased implementation plan. Importance: Maintains team morale and prevents burnout within strict constraints.

Relevant: Directly tied to client's primary goal of UX improvement and business KPIs. Importance: Ensures all efforts align with client's strategic priorities.

Time-bound: Clear deadlines for each milestone (4-8 months).

Importance: Creates urgency and enables precise resource allocation within budget limits.

Iii Cross-Functional Communication Strategy

- **Daily 15-minute stand-ups:** Focused on progress against objectives, blockers, and daily commitments using a shared digital board (e.g., Trello)
- **Bi-weekly objective review sessions:** Deep dives into metric progress with visual dashboards showing KPIs against targets
- **Collaborative documentation:** Shared Confluence space with living documents for requirements, design specs, and QA criteria
- **Role-specific workshops:** Dedicated sessions for UI/UX prototyping reviews and QA test-case validation against objectives

Iv Challenges and Mitigation Strategies

Challenge 1: Scope creep from stakeholder requests

- ✓ Mitigation 1: Implement a formal change request process requiring impact analysis on timeline/budget before approval
- ✓ Mitigation 2: Conduct bi-weekly stakeholder demos to validate direction early and manage expectations

Challenge 2: Misalignment between design and development

- ✓ Mitigation 1: Establish a "definition of ready" checklist for design handoffs including interactive prototypes and asset libraries
- ✓ Mitigation 2: Implement paired work sessions between UI designer and lead developer during critical feature implementation

Challenge 3: Performance constraints affecting UX goals

- ✓ Mitigation 1: Conduct technical feasibility spikes during planning phase for high-risk features

- ✓ Mitigation 2: Implement progressive enhancement strategy with core functionality guaranteed on all devices

Question 2: Hospital Management System Project Lifecycle

I Project Conception and Initiation

Problem Identification: Hospital administrators identified critical inefficiencies: 40% of staff time spent on manual paperwork, appointment no-show rates exceeding 30%, and medication errors due to illegible prescriptions.

Feasibility Analysis: Conducted 3-month study evaluating technical requirements (integration with existing lab systems), financial viability (ROI projection showing 200% return in 3 years), operational impact (workflow analysis across 5 departments), and legal compliance (HIPAA/GDPR requirements).

Stakeholder Alignment: Secured buy-in from medical directors, department heads, and IT leadership through workshops demonstrating how the system would reduce administrative burden by 60% and improve patient safety metrics.

Project Charter: Formalized approval document signed by hospital CEO authorizing $450K budget, 14-month timeline, and core team of eight members. Defined success criteria: 50% reduction in appointment no-shows, 99.5% system uptime, and 95% staff adoption within 6 months post-launch.

Ii Project Definition and Planning

Requirements Gathering: Conducted approximately forty five stakeholder interviews across clinical, administrative, and support staff. Created user stories for 12 key personas (e.g., "As a nurse, I need one-click access to patient medication history to prevent errors"). Validated requirements through 3 prototype testing cycles with actual hospital staff.

Scope Definition: Developed detailed Work Breakdown Structure (WBS) with 5 major deliverables: Patient Portal (mobile/web), Electronic Health Records module, Appointment Scheduling System, Pharmacy Integration, and Admin Dashboard. Explicitly excluded billing system overhaul to maintain scope boundaries.

Resource Planning: Assembled cross-functional team: 3 backend developers (Java/Spring), 2 frontend developers (React), 1 QA specialist, 1 UX designer, and 1 clinical workflow consultant. Secured dedicated test environment mirroring hospital network infrastructure.

Risk Management Plan: Identified top risks: data migration failures (mitigation: phased migration with parallel run), staff resistance (mitigation: change management program with super-users), and integration complexities (mitigation: API sandbox testing with vendor systems). Created contingency budget of 15% for high-impact risks.

Timeline Development: Created Gantt chart with 4 phases:

Discovery (3 weeks), Core Development (24 weeks), UAT & Training (6 weeks), and Go-Live Support (4 weeks). Established 8 key milestones with formal review gates requiring stakeholder sign-off before proceeding.